Local-first Coordination via Friend-to-Friend Choreographies

Vincenzo Ciancia@ ISTI-CNR Daniele Strollo@ ?

Roberto Guanciale@ KTH Emilio Tuosto@ GSSI Dragiša Žunić@ GSSI

Workshop in honour of Giangi

Pisa, 18-19 November 2026

– Motivations –

A framework for

[1] See [4, 5, 2, 1] for a formal model of local-first coordination

A framework for

friend-to-friend [7, 8]

[1] See [4, 5, 2, 1] for a formal model of local-first coordination

A framework for

friend-to-friend [7, 8]

local-first [3, 6] coordination[1]

[1] See [4, 5, 2, 1] for a formal model of local-first coordination

A framework for

friend-to-friend [7, 8]

local-first [3, 6] coordination[1]

and its (forthcoming) implementation

[1] See [4, 5, 2, 1] for a formal model of local-first coordination

Blessing and curse of clouds

The cloud, our friend!

cloud computing = fast communication infrastructures + global-scale computation facilities

The cloud, our friend!

cloud computing = fast communication infrastructures + global-scale computation facilities

- anytime-anywhere motto

The cloud, our friend!

cloud computing = fast communication infrastructures + global-scale computation facilities

- anytime-anywhere motto
- SaaS motto

The cloud, our friend!

cloud computing = fast communication infrastructures + global-scale computation facilities

- anytime-anywhere motto
- SaaS motto
- enabling dynamic reconfiguration (replace services if needed)

The cloud, our friend!

cloud computing = fast communication infrastructures + global-scale computation facilities

- anytime-anywhere motto
- SaaS motto
- enabling dynamic reconfiguration (replace services if needed)
- reduced costs for maintenance, personnel, software, hardware, $\cdots$

Except when it is not



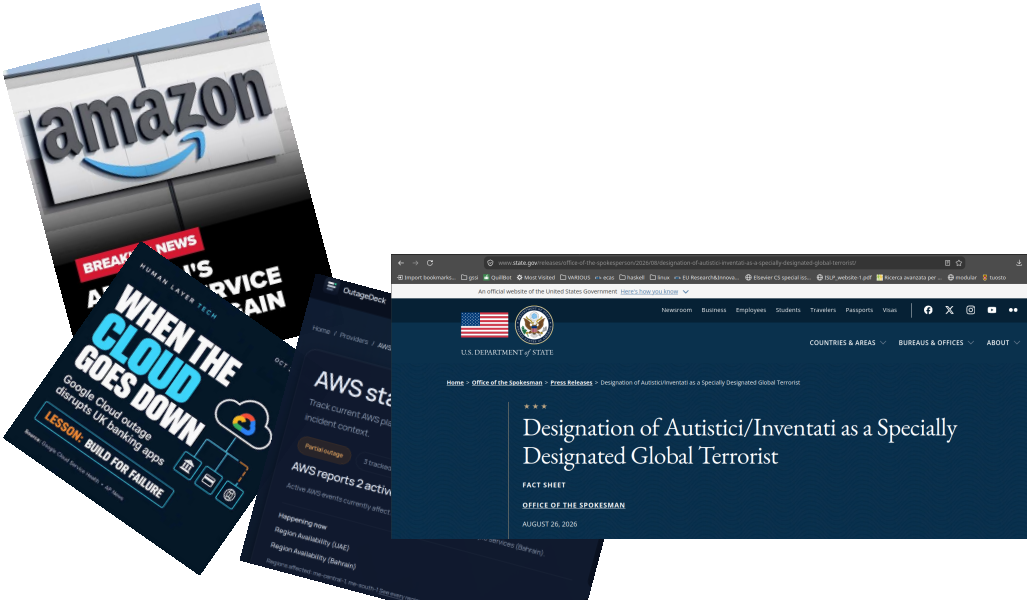
Except when it is not



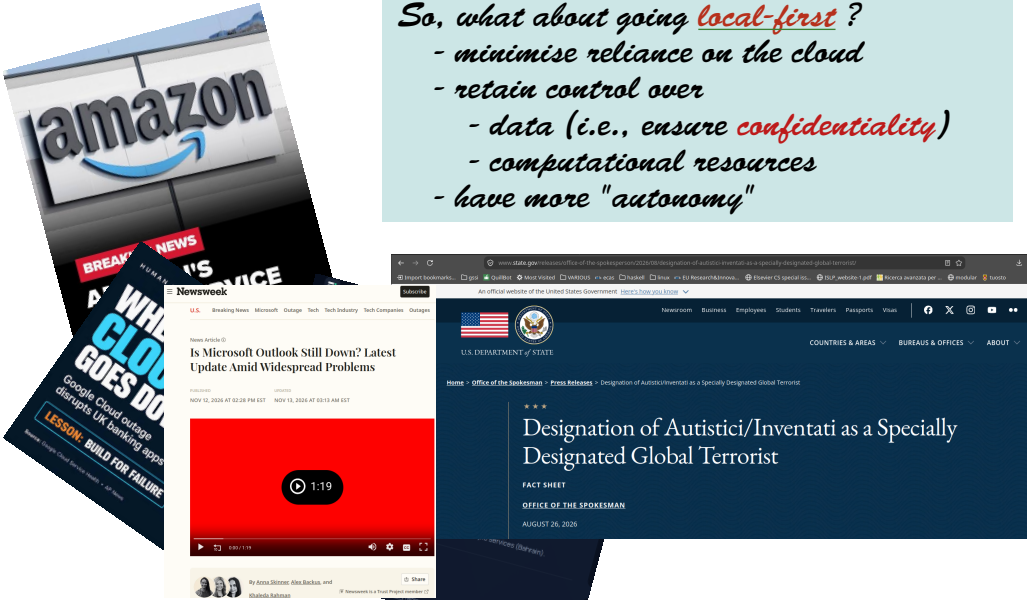
Except when it is not



Except when it is not



Except when it is not



So, what about going local-first ?

- minimise reliance on the cloud
- retain control over
  - data (i.e., ensure confidentiality)
  - computational resources
- have more "autonomy"

What if we went friend-to-friend too?

In Li & Dabek's words [7]

F2F: reliable storage in open networks

Abstract [...] We argue that a node should choose its neighbors (the nodes with which it shares resources) based on existing social relationships instead of randomly. This approach provides incentives for nodes to cooperate and results in a more stable system which, in turn, reduces the cost of maintaining data. [...]

In Popescu, Crispo & Tanenbaum words [8]

[...]Friends Team-Up and Beat the System

Abstract [...] while existing peer-to-peer architectures with similar aims attempt to build trust relationships on top of the basic, trust-agnostic, peer-to-peer overlay, Turtle takes the opposite approach, and builds its overlay on top of pre-existent trust relationships among its users. [...]

– A new choreographic model –

Our choreographic model

Syntax

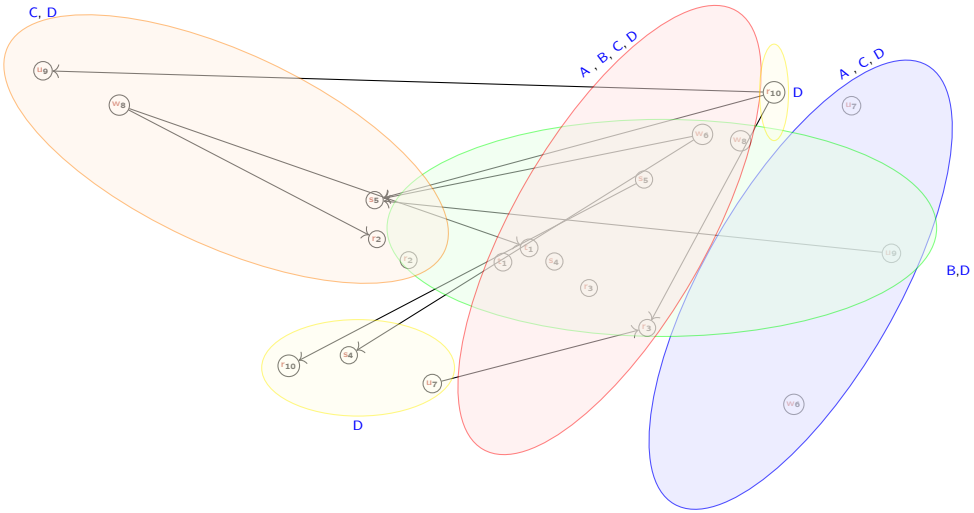
$$
\begin{aligned}
F \;::=\; & \mathsf{A} \rightsquigarrow \mathcal{B} : \mathsf{t} \\
\mid\; & F + F' \\
\mid\; & F ; F' \\
\mid\; & F \mid F' \\
\mid\; & \cdots
\end{aligned}
$$

Semantics

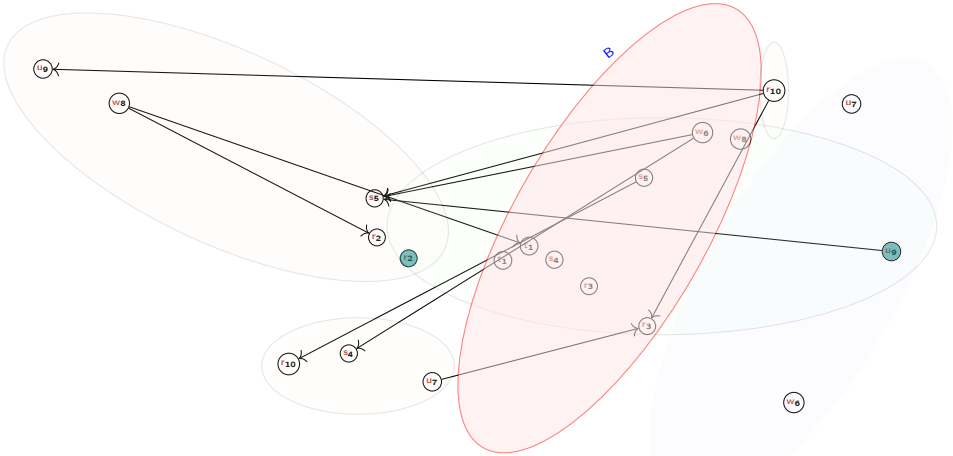
$$
\begin{aligned}
[\![\mathsf{A} \rightsquigarrow \mathcal{B} : \mathsf{t}]\!] &= \left\{ \left[\mathsf{A}!\mathcal{B}\langle \mathsf{t} \rangle \right] \right\} \\
[\![F + F']\!] &= [\![F]\!] \cup [\![F']\!] \\
[\![F ; F']\!] &= \left\{ r ;_{\mathcal{E}_r} r' \mid \cdots \right\} \\
[\![F \mid F']\!] &= \left\{ r ;_{\emptyset} r' \mid \cdots \right\} \\
[\![\cdots]\!] &= \cdots
\end{aligned}
$$

where
- $\mathcal{B}$ is a finite (possibly empty) set of A's "buddies"
- t is a "type"

Our local-first model, informally



Replication

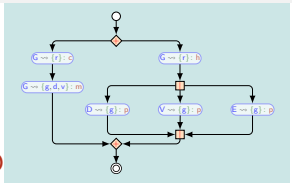


An F2F-choreography for fun

$$
\begin{aligned}
F = \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m} \\
+ \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})
\end{aligned}
$$

An F2F-choreography for fun

$$
\begin{aligned}
F = \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m} \\
+ \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p}
\end{aligned}
$$

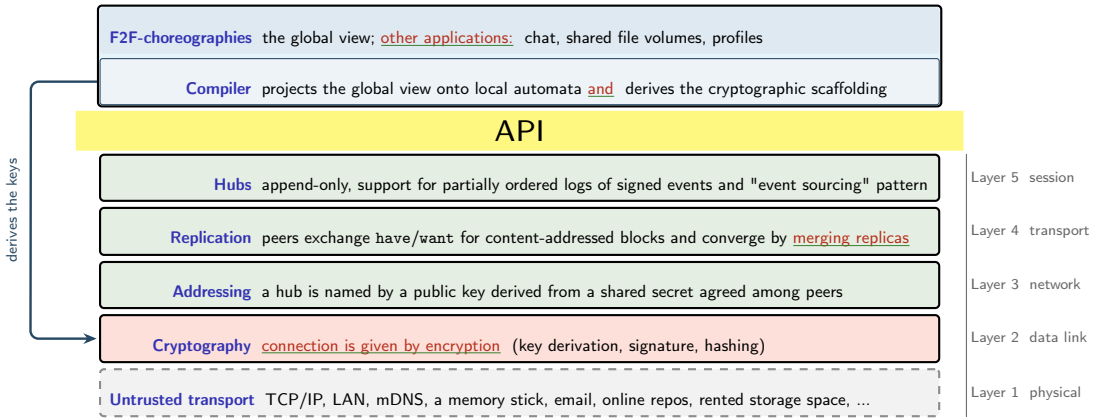


An F2F-choreography for fun

$$F = \quad \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m}$$
$$+ \quad \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})$$

$$[\![F]\!] = \left\{ \left[\begin{array}{c} \mathsf{G}!r\langle \mathsf{c} \rangle \\ \downarrow \\ \mathsf{R}!\{\mathsf{G},\mathsf{D},\mathsf{V}\}\langle \mathsf{m} \rangle \end{array} \right], \left[\begin{array}{ccc} & \mathsf{G}!r\langle \mathsf{h} \rangle & \\ \swarrow & \downarrow & \searrow \\ \mathsf{V}!\mathsf{G}\langle \mathsf{p} \rangle & \mathsf{D}!\mathsf{G}\langle \mathsf{p} \rangle & \mathsf{E}!\mathsf{G}\langle \mathsf{p} \rangle \end{array} \right] \right\}$$

F2F-choreographies on top of Nearbytes



Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \cdot E^{+} \\ H_1 & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \\ \dots & \\ H_{31} & = M \cdot G^{+} \\ H_{32} & = M \end{cases}$$

often only few needed

Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot \mathsf{G}^+ \cdot \mathsf{R}^+ \cdot \mathsf{V}^+ \cdot \mathsf{D}^+ \cdot \mathsf{E}^+ \\ H_1 & = M \cdot \mathsf{G}^+ \cdot \mathsf{R}^+ \cdot \mathsf{V}^+ \cdot \mathsf{D}^+ \\ \dots & \\ H_{31} & = M \cdot \mathsf{G}^+ \\ H_{32} & = M \end{cases}$$

often only few needed

$$\text{payload} = \text{block}_m, \quad \text{block}_\mathsf{t}, \quad \text{event}$$

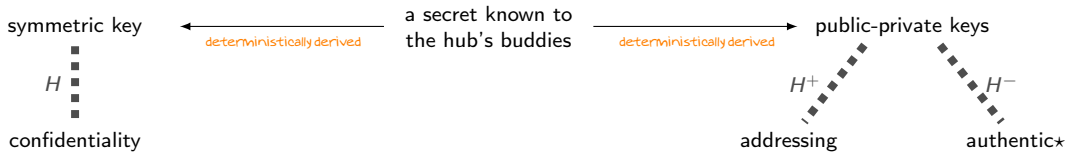
Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \cdot E^{+} \\ H_1 & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \\ \dots & \\ H_{31} & = M \cdot G^{+} \\ H_{32} & = M \end{cases}$$

often only few needed

$$\text{payload} = \text{block}_m, \quad \text{block}_t, \quad \text{event}$$



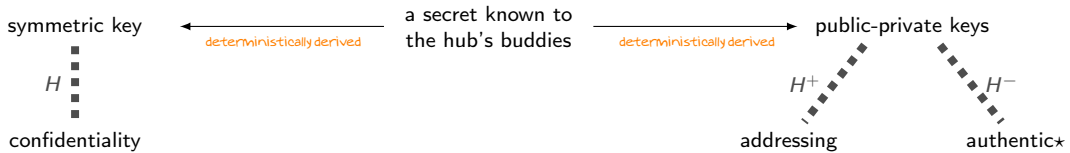
Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \cdot E^{+} \\ H_1 & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \\ \dots & \\ H_{31} & = M \cdot G^{+} \\ H_{32} & = M \end{cases}$$

often only few needed

$$\text{payload} = \text{block}_m, \quad \text{block}_t, \quad \text{event}$$



$$\text{block}_m \quad = \quad \{\, \text{msg} \,\}_H$$

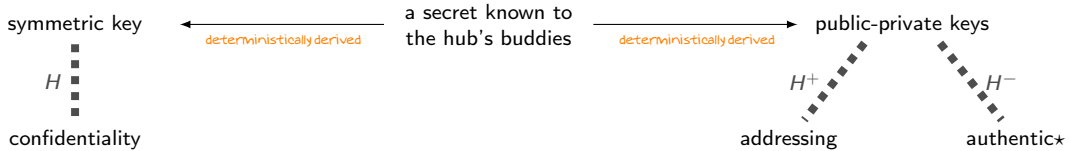
Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \cdot E^{+} \\ H_1 & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \\ \dots & \\ H_{31} & = M \cdot G^{+} \\ H_{32} & = M \end{cases}$$

often only few needed

$$\text{payload} = \text{block}_m, \quad \text{block}_t, \quad \text{event}$$



$$\begin{aligned} \text{block}_m &= \{\, \text{msg} \,\}_H \\ \text{block}_t &= \{\, type \,\}_{H^+} \end{aligned}$$

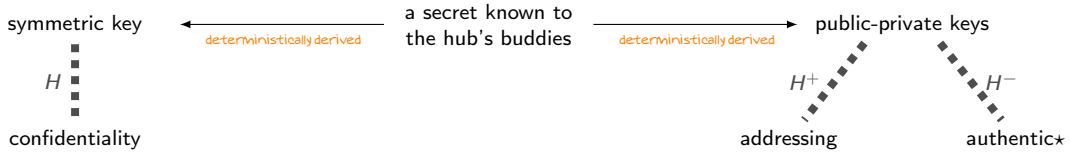
Hubs' addressing (hush-hush...no key distribution)

Buddies agree on a "master secret" M
Leverage buddies' public keys

$$\begin{cases} H & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \cdot E^{+} \\ H_1 & = M \cdot G^{+} \cdot R^{+} \cdot V^{+} \cdot D^{+} \\ \dots & \\ H_{31} & = M \cdot G^{+} \\ H_{32} & = M \end{cases}$$

often only few needed

$$\text{payload} = \text{block}_m, \quad \text{block}_t, \quad \text{event}$$



$$\begin{aligned} \text{block}_m &= \{\, \text{msg} \,\}_{H} \\ \text{block}_t &= \{\, \textit{type} \,\}_{H^{+}} \\ \text{event} &= \{\, \text{hash}(\text{ev}_1), \dots, \text{hash}(\text{ev}_k), \quad \text{hash}(\texttt{block}_m), \quad \text{hash}(\texttt{block}_t) \,\}_{H^{-}} \end{aligned}$$

Projections

Deriving buddies' expected behaviour:

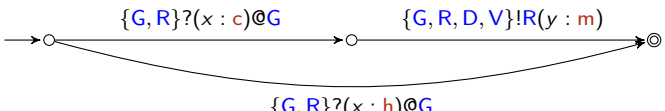
$$
\begin{aligned}
F = \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m} \\
+ \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})
\end{aligned}
$$

Projections

Deriving buddies' expected behaviour:

$$
\begin{aligned}
F = \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m} \\
+ \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})
\end{aligned}
$$

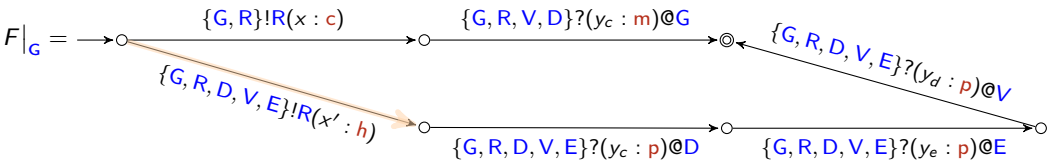
$F\big|_{\mathsf{R}} =$



Projections

Deriving buddies' expected behaviour:

$$F = \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m}$$
$$+ \quad \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})$$



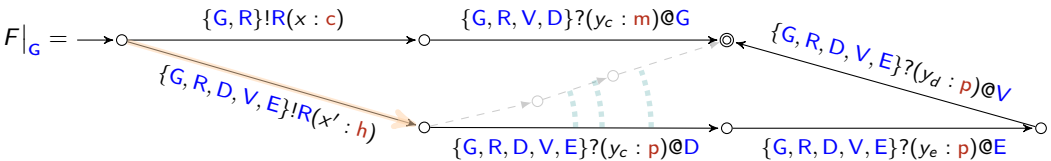
$$\left[\quad \mathsf{G}!\{\mathsf{G}, \mathsf{R}, \mathsf{D}, \mathsf{V}, \mathsf{E}\}\langle \mathsf{h} \rangle \quad\right]$$

$$e = \{\, \mathsf{G}\text{'s address} \,\}_H, \{\, \mathsf{h} \,\}_{H^+}, \{\, \mathsf{hash}(\{\, \mathsf{G}\text{'s address} \,\}_H), \mathsf{hash}(\{\, \mathsf{h} \,\}_{H^+}) \,\}_{H^-}$$

Projections

Deriving buddies' expected behaviour:

$$F = \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m}$$
$$+ \quad \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})$$



$$\left[\ \mathsf{G}!\{\mathsf{G}, \mathsf{R}, \mathsf{D}, \mathsf{V}, \mathsf{E}\}\langle \mathsf{h} \rangle\ \right]$$

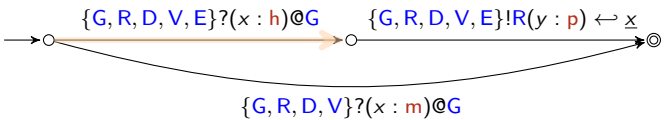
$e = \{\, \mathsf{G}\text{'s address} \,\}_H, \{\, \mathsf{h} \,\}_{H^+}, \{\, \mathsf{hash}(\{\, \mathsf{G}\text{'s address} \,\}_H), \mathsf{hash}(\{\, \mathsf{h} \,\}_{H^+}) \,\}_{H^-}$

Projections

Deriving buddies' expected behaviour:

$$
\begin{aligned}
F = \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m} \\
+ \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})
\end{aligned}
$$

$F\big|_{\mathsf{D}} =$



$$\left[\quad \mathsf{G}!\{\mathsf{G}, \mathsf{R}, \mathsf{D}, \mathsf{V}, \mathsf{E}\}\langle \mathsf{h} \rangle \quad\right]$$

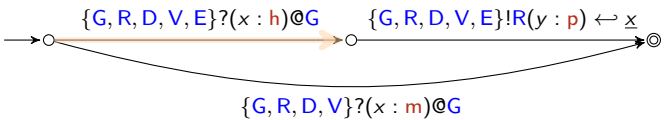
$$e = \{\, \mathsf{G}\text{'s address} \,\}_{H}, \{\, \mathsf{h} \,\}_{H^+}, \{\, \mathsf{hash}(\{\, \mathsf{G}\text{'s address} \,\}_{H}), \mathsf{hash}(\{\, \mathsf{h} \,\}_{H^+}) \,\}_{H^-}$$

Projections

Deriving buddies' expected behaviour:

$$
\begin{aligned}
F = \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{c}; \mathsf{R} \rightsquigarrow \{\mathsf{G}, \mathsf{D}, \mathsf{V}\} : \mathsf{m} \\
+ \quad & \mathsf{G} \rightsquigarrow \mathsf{R} : \mathsf{h}; (\mathsf{D} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{V} \rightsquigarrow \mathsf{G} : \mathsf{p} \mid \mathsf{E} \rightsquigarrow \mathsf{G} : \mathsf{p})
\end{aligned}
$$

$F|_{\mathsf{D}} =$



$$
\begin{bmatrix}
\mathsf{G}!\{\mathsf{G}, \mathsf{R}, \mathsf{D}, \mathsf{V}, \mathsf{E}\}\langle \mathsf{h} \rangle \\
\downarrow \\
\mathsf{D}!\{\mathsf{G}, \mathsf{R}, \mathsf{D}, \mathsf{V}, \mathsf{E}\}\langle \mathsf{p} \rangle
\end{bmatrix}
$$

$$
e = \{\, \mathsf{G}\text{'s address} \,\}_H, \{\, \mathsf{h} \,\}_{H^+}, \{\, \mathsf{hash}(\{\, \mathsf{G}\text{'s address} \,\}_H), \mathsf{hash}(\{\, \mathsf{h} \,\}_{H^+}) \,\}_{H^-}
$$

$$
\{\, \mathit{margherita} \,\}_H, \{\, \mathsf{p} \,\}_{H^+}, \{\, \mathsf{hash}(\mathsf{e}), \mathsf{hash}(\{\, \mathit{margherita} \,\}_H), \mathsf{hash}(\{\, \mathsf{p} \,\}_{H^+}) \,\}_{H^-}
$$

Executing projections in Nearbytes

Executing projections in Nearbytes

- Transitions' label univocally mapped to Nearbytes payloads
  - each output transition invokes Nearbytes API to emit the corresponding payload
  - each input is a search in the partially-ordered hub

Executing projections in Nearbytes

- Transitions' label univocally mapped to Nearbytes payloads
  - each output transition invokes Nearbytes API to emit the corresponding payload
  - each input is a search in the partially-ordered hub
- Easy realisation: generate the automata and interpret them

Executing projections in Nearbytes

- Transitions' label univocally mapped to Nearbytes payloads
  - each output transition invokes Nearbytes API to emit the corresponding payload
  - each input is a search in the partially-ordered hub
- Easy realisation: generate the automata and interpret them
- More smartly: "on-demand" generation of the automata leveraging event-notification

Executing projections in Nearbytes

- Transitions' label univocally mapped to Nearbytes payloads
  - each output transition invokes Nearbytes API to emit the corresponding payload
  - each input is a search in the partially-ordered hub
- Easy realisation: generate the automata and interpret them
- More smartly: "on-demand" generation of the automata leveraging event-notification
- Compile automata to multithreaded code (e.g., `Typescript` or `GoLang`) using Nearbytes API

– Epilogue –

"Obscured by clouds"?

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.
Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.
Magic visions stirring,
Kindled by and burning flames rise in her eyes.
The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.
She breaks the golden band.



Burning Bridges
(Pink Floyd)

Future work

Future work

- Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$\mathsf{A} \rightsquigarrow \mathsf{m}_1; \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_3 + \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_4$$

Future work

- Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$\mathsf{A} \rightsquigarrow \mathsf{m}_1; \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_3 + \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_4$$

- Consider specific notions of eventual consistency

Future work

- Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$\mathsf{A} \rightsquigarrow \mathsf{m}_1; \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_3 + \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_4$$

- Consider specific notions of eventual consistency
- Proof mechanisation

Future work

- Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$\mathsf{A} \rightsquigarrow \mathsf{m}_1; \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_3 + \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_4$$

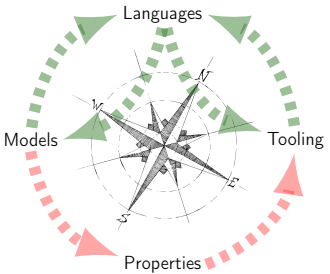
- Consider specific notions of eventual consistency
- Proof mechanisation
- Finalise the implementation
  - Broadcast encryption schemes?
  - Use higher-dimensional automata?

Future work

- Study well-formedness conditions of F2F-choreographies for "odd" protocols like

$$\mathsf{A} \rightsquigarrow \mathsf{m}_1; \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_3 + \mathsf{A} \rightsquigarrow \mathsf{m}_2; \mathsf{B} \rightsquigarrow \mathsf{m}_4$$

- Consider specific notions of eventual consistency
- Proof mechanisation
- Finalise the implementation
  - Broadcast encryption schemes?
  - Use higher-dimensional automata?



As Giangi taught us!

Thank Giangi!

References I

[1] Lucas Clorius, Florian Furbach, Roland Kuhn, Hernán Melgratti, Alceste Scalas, and Emilio Tuosto. Compositional Design, Implementation, and Verification of Swarms (Artifact).
Dagstuhl Artifacts Series, 12(1):18:1–18:3, 2026.

[2] Florian Furbach, Lucas Clorius, Roland Kuhn, Hernán C. Melgratti, Alceste Scalas, and Emilio Tuosto. Compositional Design, Implementation, and Verification of Swarms.
In Robbert Krebbers and Alexandra Silva, editors, *40th European Conference on Object-Oriented Programming, ECOOP 2026, Brussels, Belgium, June 29 - July 3, 2026*, volume 372 of *LIPIcs*, pages 7:1–7:30. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2026.

[3] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan. Local-first software: you own your data, in spite of the cloud.
In *Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software*, Onward! 2019, page 154–178, New York, NY, USA, 2019. Association for Computing Machinery.

[4] Roland Kuhn, Hernán C. Melgratti, and Emilio Tuosto. Behavioural Types for Local-First Software.
In Karim Ali and Guido Salvaneschi, editors, *37th European Conference on Object-Oriented Programming, ECOOP 2023, July 17-21, 2023, Seattle, Washington, United States*, volume 263 of *LIPIcs*, pages 15:1–15:28. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023.

[5] Roland Kuhn, Hernán C. Melgratti, and Emilio Tuosto. Behavioural Types for Local-First Software (Artifact).
Dagstuhl Artifacts Ser., 9(2):14:1–14:5, 2023.

[6] Roland Kuhn and Wesley Reisz. Local-First Cooperation: Autonomy at the Edge, Secured by Crypto, 100% Available.
https://www.infoq.com/articles/local-first-cooperation, May 2021. Last access on June 14, 2026.

[7] Jinyang Li and Frank Dabek. F2f: reliable storage in open networks.
https://iptps06.cs.ucsb.edu/papers/Li-F2F06.pdf.

References II

[8] Bogdan C. Popescu, Bruno Crispo, and Andrew S. Tanenbaum. Safe and private data sharing with turtle: Friends team-up and beat the system.
In Bruce Christianson, Bruno Crispo, James A. Malcolm, and Michael Roe, editors, *Security Protocols*, pages 213–220, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.